GAZECHIM FROID

# Table Thermodynamique R 410A

| Temperature (°C) | Liquide Pression (bar abs) | Vapeur Pression (bar abs) | Liquide Densité ($Kg/M^3$) | Vapeur Densité ($Kg/M^3$) | Liquide Enthalpie (KJ/Kg) | Vapeur Enthalpie (KJ/Kg) | Liquide Entropie (KJ/Kg-K) | Vapeur Entropie (KJ/Kg-K) |
|---|---|---|---|---|---|---|---|---|
| -70 | 0,35621 | 0,35446 | 1406,7 | 1,5584 | 101,24 | 389,25 | 0,58588 | 2,0038 |
| -69 | 0,379 | 0,37717 | 1403,6 | 1,6517 | 102,59 | 389,81 | 0,59251 | 1,9996 |
| -68 | 0,40296 | 0,40105 | 1400,6 | 1,7494 | 103,94 | 390,37 | 0,59911 | 1,9955 |
| -67 | 0,42814 | 0,42614 | 1397,6 | 1,8517 | 105,3 | 390,93 | 0,60568 | 1,9914 |
| -66 | 0,45458 | 0,45248 | 1394,6 | 1,9588 | 106,65 | 391,48 | 0,61223 | 1,9874 |
| -65 | 0,48233 | 0,48013 | 1391,5 | 2,0708 | 108 | 392,03 | 0,61874 | 1,9835 |
| -64 | 0,51143 | 0,50914 | 1388,5 | 2,1878 | 109,36 | 392,58 | 0,62523 | 1,9796 |
| -63 | 0,54194 | 0,53954 | 1385,4 | 2,31 | 110,72 | 393,13 | 0,63169 | 1,9757 |
| -62 | 0,5739 | 0,57139 | 1382,4 | 2,4376 | 112,07 | 393,68 | 0,63812 | 1,972 |
| -61 | 0,60736 | 0,60473 | 1379,3 | 2,5708 | 113,43 | 394,22 | 0,64453 | 1,9682 |
| -60 | 0,64238 | 0,63963 | 1376,2 | 2,7097 | 114,79 | 394,76 | 0,65091 | 1,9646 |
| -59 | 0,679 | 0,67612 | 1373,1 | 2,8545 | 116,15 | 395,3 | 0,65726 | 1,961 |
| -58 | 0,71728 | 0,71427 | 1370 | 3,0055 | 117,51 | 395,84 | 0,66359 | 1,9574 |
| -57 | 0,75726 | 0,75412 | 1366,9 | 3,1627 | 118,88 | 396,37 | 0,66989 | 1,9539 |
| -56 | 0,79902 | 0,79573 | 1363,8 | 3,3263 | 120,24 | 396,91 | 0,67617 | 1,9504 |
| -55 | 0,84259 | 0,83916 | 1360,7 | 3,4966 | 121,6 | 397,44 | 0,68243 | 1,947 |
| -54 | 0,88804 | 0,88445 | 1357,6 | 3,6737 | 122,97 | 397,96 | 0,68866 | 1,9436 |
| -53 | 0,93543 | 0,93168 | 1354,5 | 3,8579 | 124,34 | 398,49 | 0,69487 | 1,9403 |
| -52 | 0,9848 | 0,98089 | 1351,3 | 4,0494 | 125,7 | 399,01 | 0,70105 | 1,937 |
| -51 | 1,0362 | 1,0321 | 1348,2 | 4,2482 | 127,07 | 399,53 | 0,70722 | 1,9338 |
| -50 | 1,0898 | 1,0855 | 1345 | 4,4548 | 128,45 | 400,05 | 0,71336 | 1,9306 |
| -49 | 1,1455 | 1,141 | 1341,9 | 4,6692 | 129,82 | 400,57 | 0,71947 | 1,9275 |
| -48 | 1,2034 | 1,1988 | 1338,7 | 4,8916 | 131,19 | 401,08 | 0,72557 | 1,9244 |
| -47 | 1,2637 | 1,2588 | 1335,5 | 5,1224 | 132,57 | 401,59 | 0,73165 | 1,9213 |
| -46 | 1,3263 | 1,3212 | 1332,3 | 5,3617 | 133,95 | 402,1 | 0,7377 | 1,9183 |
| -45 | 1,3913 | 1,386 | 1329,1 | 5,6098 | 135,32 | 402,6 | 0,74374 | 1,9154 |
| -44 | 1,4588 | 1,4533 | 1325,9 | 5,8668 | 136,7 | 403,1 | 0,74975 | 1,9124 |
| -43 | 1,5288 | 1,5231 | 1322,7 | 6,1331 | 138,09 | 403,6 | 0,75575 | 1,9095 |
| -42 | 1,6015 | 1,5955 | 1319,4 | 6,4088 | 139,47 | 404,1 | 0,76173 | 1,9067 |
| -41 | 1,6769 | 1,6706 | 1316,2 | 6,6942 | 140,86 | 404,59 | 0,76768 | 1,9038 |
| -40 | 1,755 | 1,7485 | 1312,9 | 6,9896 | 142,24 | 405,08 | 0,77362 | 1,9011 |
| -39 | 1,8359 | 1,8292 | 1309,7 | 7,2952 | 143,63 | 405,57 | 0,77954 | 1,8983 |
| -38 | 1,9198 | 1,9127 | 1306,4 | 7,6112 | 145,03 | 406,05 | 0,78545 | 1,8956 |
| -37 | 2,0065 | 1,9992 | 1303,1 | 7,938 | 146,42 | 406,53 | 0,79133 | 1,8929 |
| -36 | 2,0964 | 2,0888 | 1299,8 | 8,2758 | 147,82 | 407,01 | 0,7972 | 1,8903 |
| -35 | 2,1893 | 2,1814 | 1296,5 | 8,6248 | 149,21 | 407,48 | 0,80306 | 1,8876 |
| -34 | 2,2854 | 2,2772 | 1293,1 | 8,9854 | 150,61 | 407,95 | 0,80889 | 1,885 |
| -33 | 2,3848 | 2,3762 | 1289,8 | 9,3578 | 152,02 | 408,42 | 0,81471 | 1,8825 |
| -32 | 2,4875 | 2,4786 | 1286,4 | 9,7423 | 153,42 | 408,88 | 0,82052 | 1,88 |
| -31 | 2,5935 | 2,5843 | 1283,1 | 10,139 | 154,83 | 409,34 | 0,82631 | 1,8775 |
| -30 | 2,7031 | 2,6935 | 1279,7 | 10,549 | 156,24 | 409,8 | 0,83208 | 1,875 |
| -29 | 2,8162 | 2,8062 | 1276,3 | 10,972 | 157,65 | 410,25 | 0,83784 | 1,8725 |
| -28 | 2,9329 | 2,9226 | 1272,9 | 11,408 | 159,06 | 410,7 | 0,84358 | 1,8701 |
| -27 | 3,0534 | 3,0426 | 1269,4 | 11,857 | 160,48 | 411,14 | 0,84931 | 1,8677 |
| -26 | 3,1776 | 3,1665 | 1266 | 12,321 | 161,9 | 411,58 | 0,85503 | 1,8654 |

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GAZECHIM FROID

# Table Thermodynamique R 410A

| Temperature (°C) | Liquide Pression (bar abs) | Vapeur Pression (bar abs) | Liquide Densité ($Kg/M^3$) | Vapeur Densité ($Kg/M^3$) | Liquide Enthalpie (KJ/Kg) | Vapeur Enthalpie (KJ/Kg) | Liquide Entropie (KJ/Kg-K) | Vapeur Entropie (KJ/Kg-K) |
|---|---|---|---|---|---|---|---|---|
| -25 | 3,3057 | 3,2941 | 1262,5 | 12,799 | 163,32 | 412,02 | 0,86074 | 1,863 |
| -24 | 3,4377 | 3,4257 | 1259,1 | 13,292 | 164,75 | 412,45 | 0,86643 | 1,8607 |
| -23 | 3,5738 | 3,5613 | 1255,6 | 13,8 | 166,18 | 412,88 | 0,87211 | 1,8584 |
| -22 | 3,7139 | 3,701 | 1252,1 | 14,323 | 167,61 | 413,31 | 0,87777 | 1,8562 |
| -21 | 3,8582 | 3,8449 | 1248,6 | 14,861 | 169,04 | 413,73 | 0,88343 | 1,8539 |
| -20 | 4,0069 | 3,993 | 1245 | 15,416 | 170,48 | 414,14 | 0,88907 | 1,8517 |
| -19 | 4,1598 | 4,1455 | 1241,5 | 15,988 | 171,92 | 414,55 | 0,8947 | 1,8495 |
| -18 | 4,3172 | 4,3024 | 1237,9 | 16,576 | 173,37 | 414,96 | 0,90032 | 1,8473 |
| -17 | 4,4791 | 4,4638 | 1234,3 | 17,182 | 174,81 | 415,36 | 0,90593 | 1,8451 |
| -16 | 4,6456 | 4,6298 | 1230,7 | 17,805 | 176,26 | 415,76 | 0,91153 | 1,843 |
| -15 | 4,8168 | 4,8004 | 1227,1 | 18,447 | 177,72 | 416,15 | 0,91712 | 1,8408 |
| -14 | 4,9927 | 4,9758 | 1223,4 | 19,107 | 179,18 | 416,54 | 0,9227 | 1,8387 |
| -13 | 5,1735 | 5,1561 | 1219,7 | 19,786 | 180,64 | 416,92 | 0,92827 | 1,8366 |
| -12 | 5,3593 | 5,3412 | 1216 | 20,485 | 182,1 | 417,3 | 0,93383 | 1,8345 |
| -11 | 5,5501 | 5,5314 | 1212,3 | 21,203 | 183,57 | 417,67 | 0,93938 | 1,8325 |
| -10 | 5,746 | 5,7268 | 1208,6 | 21,942 | 185,04 | 418,04 | 0,94493 | 1,8304 |
| -9 | 5,9472 | 5,9273 | 1204,8 | 22,702 | 186,52 | 418,4 | 0,95047 | 1,8284 |
| -8 | 6,1536 | 6,1331 | 1201,1 | 23,484 | 188 | 418,76 | 0,956 | 1,8264 |
| -7 | 6,3654 | 6,3442 | 1197,3 | 24,287 | 189,48 | 419,11 | 0,96152 | 1,8244 |
| -6 | 6,5828 | 6,5609 | 1193,4 | 25,113 | 190,97 | 419,45 | 0,96703 | 1,8224 |
| -5 | 6,8057 | 6,7831 | 1189,6 | 25,962 | 192,47 | 419,79 | 0,97254 | 1,8204 |
| -4 | 7,0342 | 7,011 | 1185,7 | 26,835 | 193,96 | 420,12 | 0,97804 | 1,8184 |
| -3 | 7,2686 | 7,2446 | 1181,8 | 27,732 | 195,47 | 420,45 | 0,98354 | 1,8164 |
| -2 | 7,5087 | 7,484 | 1177,9 | 28,654 | 196,97 | 420,77 | 0,98903 | 1,8145 |
| -1 | 7,7549 | 7,7294 | 1173,9 | 29,601 | 198,48 | 421,08 | 0,99452 | 1,8125 |
| 0 | 8,0071 | 7,9808 | 1169,9 | 30,575 | 200 | 421,39 | 1 | 1,8106 |
| 1 | 8,2654 | 8,2384 | 1165,9 | 31,575 | 201,52 | 421,69 | 1,0055 | 1,8086 |
| 2 | 8,5299 | 8,5021 | 1161,9 | 32,603 | 203,05 | 421,98 | 1,011 | 1,8067 |
| 3 | 8,8008 | 8,7722 | 1157,8 | 33,659 | 204,58 | 422,27 | 1,0164 | 1,8048 |
| 4 | 9,0782 | 9,0487 | 1153,7 | 34,744 | 206,11 | 422,54 | 1,0219 | 1,8029 |
| 5 | 9,3621 | 9,3318 | 1149,6 | 35,859 | 207,66 | 422,81 | 1,0274 | 1,801 |
| 6 | 9,6526 | 9,6214 | 1145,4 | 37,005 | 209,2 | 423,08 | 1,0328 | 1,7991 |
| 7 | 9,9498 | 9,9178 | 1141,2 | 38,182 | 210,76 | 423,33 | 1,0383 | 1,7972 |
| 8 | 10,254 | 10,221 | 1137 | 39,392 | 212,31 | 423,58 | 1,0437 | 1,7952 |
| 9 | 10,565 | 10,531 | 1132,8 | 40,634 | 213,88 | 423,82 | 1,0492 | 1,7933 |
| 10 | 10,883 | 10,848 | 1128,5 | 41,911 | 215,45 | 424,05 | 1,0546 | 1,7914 |
| 11 | 11,208 | 11,172 | 1124,1 | 43,223 | 217,03 | 424,27 | 1,0601 | 1,7895 |
| 12 | 11,541 | 11,504 | 1119,7 | 44,571 | 218,61 | 424,48 | 1,0656 | 1,7876 |
| 13 | 11,881 | 11,843 | 1115,3 | 45,957 | 220,2 | 424,69 | 1,071 | 1,7857 |
| 14 | 12,228 | 12,189 | 1110,9 | 47,38 | 221,79 | 424,88 | 1,0765 | 1,7838 |
| 15 | 12,583 | 12,543 | 1106,4 | 48,844 | 223,4 | 425,07 | 1,0819 | 1,7819 |
| 16 | 12,945 | 12,904 | 1101,9 | 50,348 | 225,01 | 425,24 | 1,0874 | 1,78 |
| 17 | 13,316 | 13,274 | 1097,3 | 51,894 | 226,62 | 425,41 | 1,0929 | 1,7781 |
| 18 | 13,694 | 13,651 | 1092,7 | 53,483 | 228,25 | 425,56 | 1,0983 | 1,7761 |
| 19 | 14,08 | 14,036 | 1088 | 55,117 | 229,88 | 425,7 | 1,1038 | 1,7742 |

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GAZECHIM FROID

# Table Thermodynamique R 410A

| Temperature (°C) | Liquide Pression (bar abs) | Vapeur Pression (bar abs) | Liquide Densité ($Kg/M^3$) | Vapeur Densité ($Kg/M^3$) | Liquide Enthalpie (KJ/Kg) | Vapeur Enthalpie (KJ/Kg) | Liquide Entropie (KJ/Kg-K) | Vapeur Entropie (KJ/Kg-K) |
|---|---|---|---|---|---|---|---|---|
| 20 | 14,475 | 14,429 | 1083,3 | 56,798 | 231,52 | 425,83 | 1,1093 | 1,7722 |
| 21 | 14,877 | 14,831 | 1078,5 | 58,525 | 233,16 | 425,95 | 1,1148 | 1,7703 |
| 22 | 15,288 | 15,241 | 1073,7 | 60,303 | 234,82 | 426,06 | 1,1202 | 1,7683 |
| 23 | 15,708 | 15,659 | 1068,8 | 62,131 | 236,48 | 426,16 | 1,1257 | 1,7663 |
| 24 | 16,136 | 16,086 | 1063,9 | 64,012 | 238,15 | 426,24 | 1,1312 | 1,7643 |
| 25 | 16,572 | 16,521 | 1058,9 | 65,948 | 239,84 | 426,31 | 1,1368 | 1,7623 |
| 26 | 17,018 | 16,966 | 1053,9 | 67,94 | 241,53 | 426,37 | 1,1423 | 1,7603 |
| 27 | 17,473 | 17,419 | 1048,8 | 69,991 | 243,22 | 426,41 | 1,1478 | 1,7582 |
| 28 | 17,936 | 17,882 | 1043,6 | 72,103 | 244,93 | 426,44 | 1,1533 | 1,7561 |
| 29 | 18,409 | 18,353 | 1038,4 | 74,279 | 246,65 | 426,46 | 1,1589 | 1,7541 |
| 30 | 18,891 | 18,834 | 1033,1 | 76,52 | 248,38 | 426,46 | 1,1644 | 1,7519 |
| 31 | 19,383 | 19,324 | 1027,7 | 78,83 | 250,12 | 426,44 | 1,17 | 1,7498 |
| 32 | 19,884 | 19,824 | 1022,3 | 81,211 | 251,87 | 426,41 | 1,1756 | 1,7476 |
| 33 | 20,395 | 20,334 | 1016,7 | 83,666 | 253,63 | 426,35 | 1,1812 | 1,7454 |
| 34 | 20,916 | 20,854 | 1011,2 | 86,199 | 255,41 | 426,28 | 1,1868 | 1,7432 |
| 35 | 21,447 | 21,383 | 1005,5 | 88,813 | 257,19 | 426,2 | 1,1924 | 1,741 |
| 36 | 21,988 | 21,923 | 999,71 | 91,511 | 258,99 | 426,09 | 1,1981 | 1,7387 |
| 37 | 22,539 | 22,473 | 993,86 | 94,298 | 260,8 | 425,96 | 1,2038 | 1,7364 |
| 38 | 23,101 | 23,033 | 987,91 | 97,179 | 262,63 | 425,81 | 1,2095 | 1,734 |
| 39 | 23,673 | 23,604 | 981,87 | 100,16 | 264,47 | 425,64 | 1,2152 | 1,7316 |
| 40 | 24,256 | 24,186 | 975,72 | 103,24 | 266,32 | 425,45 | 1,2209 | 1,7291 |
| 41 | 24,85 | 24,779 | 969,46 | 106,42 | 268,19 | 425,23 | 1,2267 | 1,7266 |
| 42 | 25,455 | 25,382 | 963,08 | 109,73 | 270,08 | 424,98 | 1,2325 | 1,7241 |
| 43 | 26,071 | 25,997 | 956,58 | 113,15 | 271,99 | 424,71 | 1,2383 | 1,7215 |
| 44 | 26,699 | 26,624 | 949,95 | 116,7 | 273,91 | 424,41 | 1,2442 | 1,7188 |
| 45 | 27,338 | 27,261 | 943,19 | 120,38 | 275,85 | 424,08 | 1,2501 | 1,7161 |
| 46 | 27,988 | 27,911 | 936,28 | 124,21 | 277,81 | 423,72 | 1,256 | 1,7133 |
| 47 | 28,651 | 28,572 | 929,21 | 128,19 | 279,79 | 423,33 | 1,262 | 1,7104 |
| 48 | 29,325 | 29,246 | 921,98 | 132,33 | 281,8 | 422,9 | 1,268 | 1,7075 |
| 49 | 30,012 | 29,932 | 914,57 | 136,64 | 283,83 | 422,43 | 1,2741 | 1,7044 |
| 50 | 30,711 | 30,63 | 906,97 | 141,15 | 285,88 | 421,92 | 1,2802 | 1,7013 |
| 51 | 31,422 | 31,341 | 899,16 | 145,85 | 287,97 | 421,37 | 1,2864 | 1,698 |
| 52 | 32,147 | 32,065 | 891,12 | 150,77 | 290,08 | 420,77 | 1,2927 | 1,6947 |
| 53 | 32,884 | 32,802 | 882,84 | 155,93 | 292,23 | 420,12 | 1,299 | 1,6912 |
| 54 | 33,635 | 33,552 | 874,29 | 161,35 | 294,41 | 419,41 | 1,3054 | 1,6876 |
| 55 | 34,398 | 34,316 | 865,44 | 167,04 | 296,63 | 418,65 | 1,3119 | 1,6839 |
| 56 | 35,176 | 35,094 | 856,26 | 173,05 | 298,89 | 417,82 | 1,3186 | 1,6799 |
| 57 | 35,967 | 35,886 | 846,72 | 179,4 | 301,2 | 416,92 | 1,3253 | 1,6758 |
| 58 | 36,773 | 36,692 | 836,76 | 186,14 | 303,57 | 415,94 | 1,3322 | 1,6715 |
| 59 | 37,592 | 37,513 | 826,34 | 193,3 | 305,99 | 414,87 | 1,3392 | 1,667 |
| 60 | 38,426 | 38,348 | 815,37 | 200,95 | 308,48 | 413,7 | 1,3464 | 1,6622 |
| 61 | 39,276 | 39,199 | 803,78 | 209,17 | 311,06 | 412,41 | 1,3537 | 1,6571 |
| 62 | 40,14 | 40,066 | 791,46 | 218,03 | 313,72 | 410,99 | 1,3614 | 1,6517 |
| 63 | 41,019 | 40,949 | 778,25 | 227,66 | 316,5 | 409,42 | 1,3693 | 1,6458 |
| 64 | 41,915 | 41,848 | 763,97 | 238,21 | 319,41 | 407,67 | 1,3776 | 1,6395 |

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GAZECHIM FROID

## Table Thermodynamique R 410A

| | Liquide | Vapeur | Liquide | Vapeur | Liquide | Vapeur | Liquide | Vapeur |
|---|---|---|---|---|---|---|---|---|
| Temperature | Pression | Pression | Densité | Densité | Enthalpie | Enthalpie | Entropie | Entropie |
| (°C) | (bar abs) | (bar abs) | ($Kg/M^3$) | ($Kg/M^3$) | (KJ/Kg) | (KJ/Kg) | (KJ/Kg-K) | (KJ/Kg-K) |
| 65 | 42,826 | 42,763 | 748,34 | 249,89 | 322,49 | 405,69 | 1,3864 | 1,6325 |
| 66 | 43,754 | 43,697 | 730,95 | 263,01 | 325,78 | 403,43 | 1,3957 | 1,6247 |
| 67 | 44,699 | 44,648 | 711,18 | 278,04 | 329,38 | 400,8 | 1,4059 | 1,6159 |
| 68 | 45,661 | 45,618 | 687,93 | 295,74 | 333,4 | 397,65 | 1,4174 | 1,6057 |
| 69 | 46,641 | 46,607 | 659,03 | 317,59 | 338,15 | 393,73 | 1,4308 | 1,5933 |
| 70 | 47,639 | 47,617 | 618,88 | 347,08 | 344,34 | 388,39 | 1,4484 | 1,5768 |

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